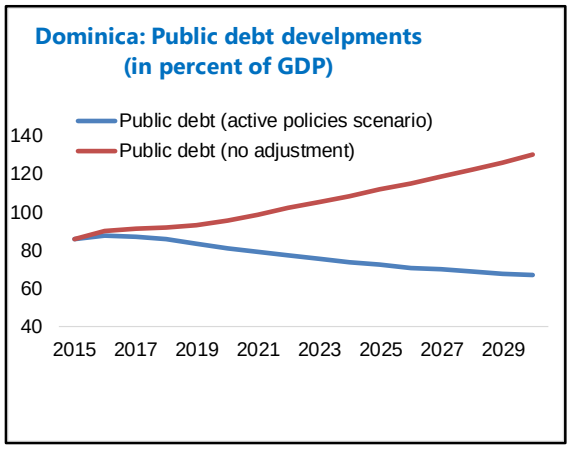
Annex I. Debt Sustainability Analysis[1]

Dominica continues to be at high risk of debt distress. The fiscal adjustment program being pursued by the authorities (active policies scenario) would put debt on a sustainable path but above the ECCU regional debt target. Customized scenarios (assuming no further fiscal consolidation, lower donor grants, lower ECP revenues and a hurricane shock in 2018) are used to inform the external risk rating, given the challenging medium-term fiscal consolidation. Should the planned fiscal consolidation take hold and the debt outlook move firmly along the sustainable path projected under the active policies scenario, an improvement of the risk rating could be considered in a future assessment.

1. Dominica's overall public debt has been increasing steadily since 2010, reflecting the weakening fiscal position. As of end-FY 2014/15, the stock of public sector debt (central government and rest of the public sector, henceforth public debt) is estimated to be around 84 percent of GDP. Over 70 percent of the public debt is external, owed mainly to multilateral creditors and the remaining is held domestically, mostly by commercial banks and other financial institutions. The debt of the rest of public sector (state owned enterprises and Petrocaribe) is about 21 percent of GDP.

2. In accordance with the fiscal consolidation measures discussed under the RCF, public debt is assessed to be sustainable. The authorities have committed to a gradual fiscal consolidation over the next five years, supported by comprehensive reforms to improve revenue collection, broaden the tax base, and contain current expenditures. They have adopted the initial fiscal measures in the RCF, and have undertaken new measures to further strengthen the fiscal position. If the fiscal consolidation plan is fully implemented, the debt outlook would improve significantly. This consolidation includes a provision of 1.5 percent of GDP per year to lower debt and to build buffers to address future natural disasters, providing a degree of resilience to the fiscal consolidation strategy. However, if the authorities did not take any additional fiscal consolidation measures, public debt would be on an upward trajectory and become unsustainable – reaching 130 percent of GDP by 2030 (text chart, no adjustment scenario).



The main differences relative to the DSA in the RCF Staff Report are as follows:

[1] This DSA has been cleared by Mr. Bob Traa (IMF) and Mr. John Panzer (World Bank).

- The growth projections for 2015 have been revised downwards as the latest data reflect a more protracted negative impact of the storm on growth.
- The debt projections now include the expected fiscal costs of future natural disasters. An annual average reconstruction cost of 1.5 percentage point of GDP is included as an expenditure on account of possible costs for reconstruction in the event of other natural disasters. This augments the deficit and worsens debt dynamics relative to the RCF. In the RCF we considered the need for an additional primary surplus of 1.5 percentage point of GDP to build a cushion for natural disasters, but the cost of natural disasters was not included in the deficit and debt projections.

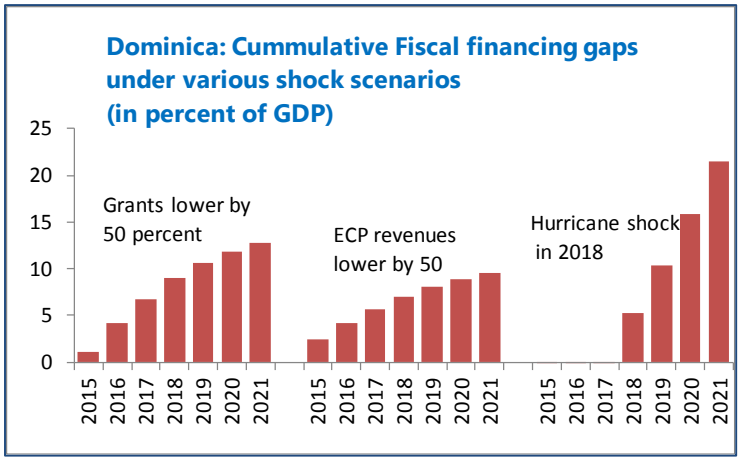
- The public debt for FY 2015/16 now incorporates the Petrocaribe debt estimated at 9 percent of GDP. The estimation is based on the Petrocaribe loan agreement. It should be noted, however, that Petrocaribe debt service payments have been cancelled or rescheduled in the last three years, a pattern that may continue into the coming years. However, the revised DSA includes these obligations for prudence given the uncertainty about the continuation of this practice.

Debt Related to the Petrocaribe Agreement (percent of GDP)

	Est.	Proj.					
	2015	2016	2017	2018	2019	2020	2021
Imports Under Petrocaribe	2.0	1.6	1.8	1.9	1.9	1.9	1.9
of which: financed with borrowing	0.8	0.4	0.5	0.6	0.6	0.6	0.6
Debt Under Petrocaribe	8.8	8.7	8.5	8.3	8.1	7.9	7.7

- On the positive side, the authorities' fiscal program is over-performing slightly (0.7% of GDP) based on the fiscal measures undertaken in FY 2015/16 and those proposed in the budget for FY 2016/17. In addition, the recent parametric reforms undertaken by the Dominica Social Security program (higher employee contributions and increase in retirement age) also improve public debt dynamics.

3. Although in the active scenario public debt is sustainable, the large number and potential size of risks merit the classification of Dominica as "high risk of debt distress". All the external debt and debt service ratios are below their relevant thresholds in the active scenario, which assumes a full implementation of the fiscal consolidation plan. However,



under the customized scenarios several thresholds are breached. One customized scenario (no adjustment) assumes that fiscal performance follows passively from the current set of policies in place and no additional measures are adopted. In the customized scenario, the average annual external debt service obligations are estimated around US$39.3 million (7.3% of GDP) during the next three fiscal years (FY2015-17), which could pose a financing risk. Staff simulations indicate that the materialization of other risks would also breach the debt thresholds and render unsustainable debt dynamics as shown in Figures A1 and A2 for other customized scenarios. This is particularly plausible in the cases of (i) a reduction of grants (to 50 percent of the active scenario); (ii) lower ECP revenues (to 50 percent of the active scenario); and (iii) the occurrence of a natural disaster in 2018. It should be noted that these events are largely or completely outside the government control.

4. The precarious debt sustainability prospects, even if the fiscal consolidation plan was fully implemented, prompted the government to seek debt relief from bilateral creditors. In light of the high debt burden, the large fiscal costs of the reconstruction of public infrastructure after the tropical storm, and its protracted effects on output, the authorities have formally requested debt relief from bilateral creditors as part of their fiscal consolidation strategy – including from the Paris Club and also on Petrocaribe debt service flows. Should any of the risks to debt sustainability mentioned above materialize, there would be no realistic amount of fiscal consolidation that could cover the resulting financing gap. In that case, the insuficient revenues would force a compression government spending which would most likelly affect capital expenditures and reduce growth prospects.

5. As a result, the public and publicly guaranteed (PPG) external debt risk rating remains high. This assessment is based on the risks reflected in the customized scenarios and the significant risks explained above, which carry significant weight in this assessment. Under the customized scenario, the debt path takes an increasing trajectory and breaches two solvency thresholds (Figure A1).

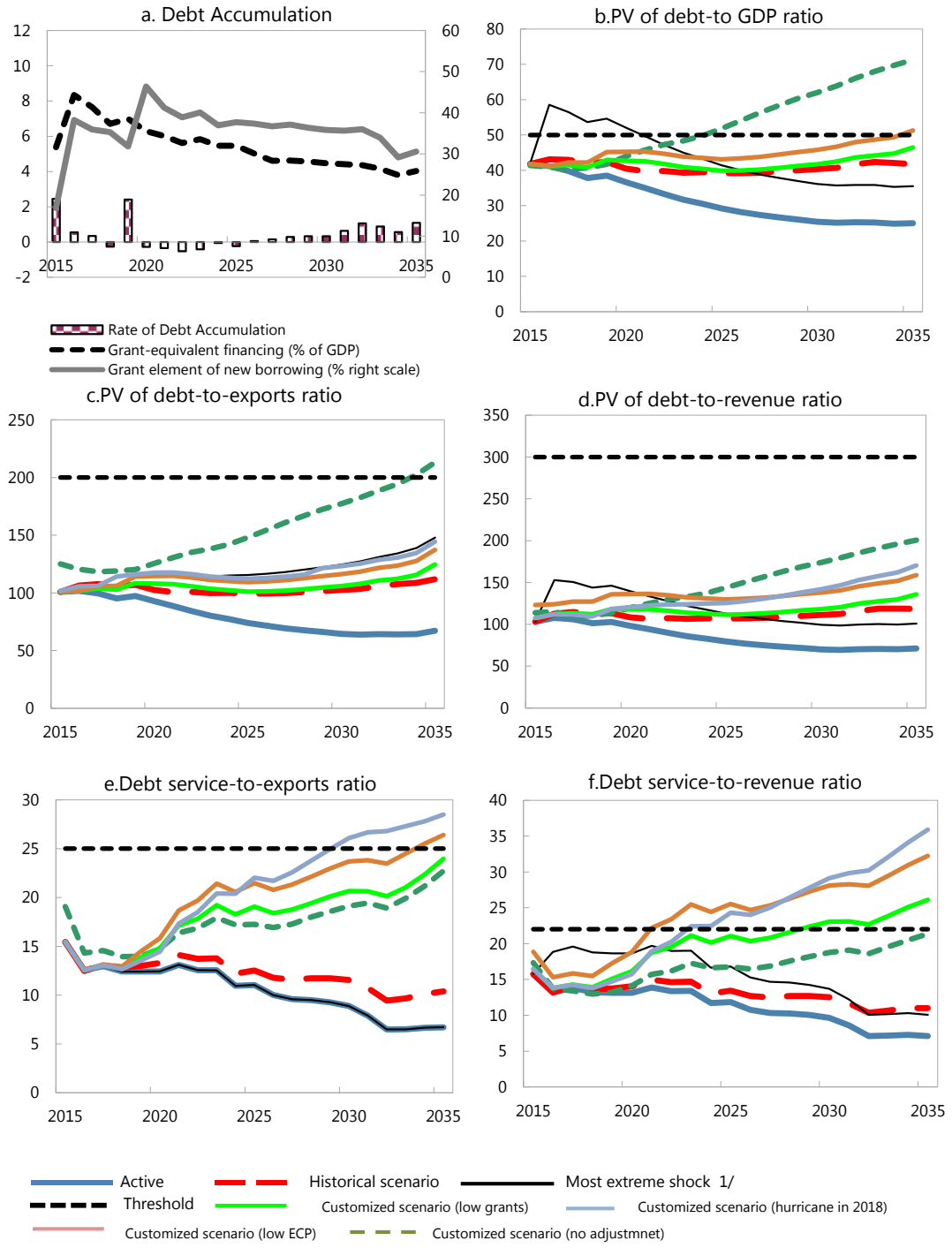
6. The authorities concurred with staff that Dominica remains at high risk of debt distress. They agreed that the debt dynamics could easily become unsustainable even if the fiscal consolidation committed in the RCF is fully implemented. They explained that the expenditure in rehabilitation and reconstruction after tropical storm Erika was largely non-optional, as required to restore normal functioning of the economy and the provision of basic public services. Given this situation, they expressed concern that shortages in grant revenues relative to current expectations or in other sources of revenue, or other natural disaster events, could result in higher debt issuance (mostly concessional) and a breaching of the debt thresholds. However, they explained that they remain fully committed to fiscal sustainability and to this end they are also pursuing debt relief from bilateral partners as part of their debt consolidation strategy.

Conclusion

7. The risk rating assigned to PPG external debt is high. Under a customized scenario (without further fiscal adjustment), debt would exceed two solvency thresholds in the medium term.

The other customized scenarios (with lower grants, lower ECP and hurricane shock) show breaches of debt service thresholds. Moreover, given the challenges associated with fiscal consolidation in the wake of storm Erika, the customized scenarios are attached significant weight in the risk assessment. Nevertheless, the authorities have expressed commitment to the fiscal adjustment program (active scenario) that would put public debt on a sustainable trajectory. As the planned fiscal consolidation takes hold, further increasing confidence in the active policies scenario, an upgrade of the risk rating could be considered in a future assessment.

Figure A1. Dominica: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2015-35 (Active Scenario) 1/



1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a Customized scenario (no adjustment) shock; in c. to a Customized scenario (no adjustment) shock; in d. to a Customized scenario (no adjustment) shock; in e. to a Customized scenario (no adjustment) shock and in figure f. to a Customized scenario (no adjustment) shock

Table A1. Dominica: External Debt Sustainability Framework, Active Scenario, 2012-35 1/
(In percent of GDP)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2012	2013	2014			2015	2016	2017	2018	2019	2020	2015-2020 Average	2025	2035	2021-2035 Average
External debt (nominal) 1/	**74.7**	**77.7**	**85.8**			**92.6**	**91.0**	**90.2**	**87.2**	**83.7**	**80.1**		**69.1**	**53.5**	
of which: public and publicly guaranteed (PPG)	52.6	52.6	53.4			56.6	56.4	55.8	53.2	51.8	50.3		45.8	37.5	
Change in external debt	1.9	3.0	8.1			6.8	-1.6	-0.8	-3.0	-3.5	-3.6		-1.8	-3.4	
Identified net debt-creating flows	7.6	7.1	3.3			9.2	2.1	2.9	3.3	5.8	4.7		3.8	3.0	
Non-interest current account deficit	**11.7**	**13.8**	**6.6**	**13.7**	**5.2**	**8.5**	**6.5**	**9.3**	**10.3**	**11.0**	**8.5**		**7.2**	**8.6**	7.5
Deficit in balance of goods and services	14.3	11.1	11.4			11.6	13.1	13.8	12.6	10.3	9.2		9.6	9.0	
Exports	35.5	39.8	41.7			41.4	40.6	40.0	39.7	39.6	39.5		39.5	37.3	
Imports	49.8	51.0	53.1			53.0	53.7	53.8	52.3	49.9	48.7		49.1	46.3	
Net current transfers (negative = inflow)	-3.7	-4.4	-5.3	**-4.3**	**0.7**	-5.2	-4.8	-4.7	-4.7	-4.7	-4.7		-4.8	-4.5	-4.7
of which: official	-0.6	-1.1	-1.1			-0.9	-0.6	-0.6	-0.6	-0.6	-0.6		-0.6	-0.5	
Other current account flows (negative = net inflow)	1.2	7.1	0.4			2.1	-1.9	0.2	2.5	5.4	4.1		2.3	4.0	
Net FDI (negative = inflow)	**-6.3**	**-6.4**	**-6.9**	**-9.7**	**2.6**	**-5.9**	**-5.8**	**-6.5**	**-7.1**	**-5.4**	**-4.2**		**-4.6**	**-4.5**	-5.0
Endogenous debt dynamics 2/	**2.1**	**-0.3**	**3.6**			**6.6**	**1.5**	**0.1**	**0.2**	**0.2**	**0.3**		**1.2**	**-1.1**	
Contribution from nominal interest rate	2.5	2.6	3.1			3.2	2.6	2.5	2.5	2.3	2.1		2.4	0.9	
Contribution from real GDP growth	0.9	-0.4	-3.0			3.4	-1.1	-2.5	-2.3	-2.1	-1.8		-1.1	0.0	
Contribution from price and exchange rate changes	-1.3	-2.4	3.6			-1.8	-1.4	-1.4	-1.6	-1.7	-1.5		-1.3	-2.0	
Residual (3-4) 3/	**-5.7**	**-4.1**	**4.8**	**-3.9**		**-2.4**	**-3.7**	**-3.7**	**-6.3**	**-9.3**	**-8.3**		**-5.6**	**-6.4**	
of which: Capital transfers						4.8	6.3	6.5	6.1	5.3	4.8		3.9	3.6	
of which: Commercial Banks and other private flows						-1.5	-0.9	0.5	-0.7	0.6	1.2		-0.6	0.0	
of which: exceptional financing	0.0	0.0	0.0			1.7	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	70.9			77.8	75.8	75.4	73.0	71.5	67.5		53.2	41.0	
In percent of exports	...	...	170.1			187.7	186.9	188.6	183.6	180.5	170.8		134.7	109.9	
PV of PPG external debt	**...**	**...**	**38.4**			**41.8**	**41.2**	**41.0**	**38.9**	**39.5**	**37.7**		**29.8**	**25.1**	
In percent of exports	**...**	**...**	**92.2**			**100.9**	**101.6**	**102.5**	**98.0**	**99.9**	**95.4**		**75.5**	**67.1**	
In percent of government revenues	**...**	**...**	**109.9**			**103.0**	**107.6**	**109.2**	**104.5**	**105.7**	**100.9**		**81.1**	**71.2**	
Debt service-to-exports ratio (in percent)	**16.5**	**13.0**	**16.6**			**19.7**	**16.9**	**17.4**	**16.9**	**16.7**	**16.7**		**15.0**	**7.7**	
PPG debt service-to-exports ratio (in percent)	**11.6**	**8.5**	**12.5**			**15.4**	**12.5**	**13.0**	**12.5**	**12.5**	**12.5**		**11.3**	**6.7**	
PPG debt service-to-revenue ratio (in percent)	**11.2**	**9.5**	**14.9**			**15.8**	**13.3**	**13.9**	**13.4**	**13.3**	**13.3**		**12.2**	**7.1**	
Total gross financing need (Billions of U.S. dollars)	0.1	0.1	0.0			0.1	0.0	0.1	0.1	0.1	0.1		0.1	0.1	
Non-interest current account deficit that stabilizes debt ratio	9.9	10.8	-1.6			1.7	8.1	10.1	13.3	14.5	12.2		9.1	12.0	
Key macroeconomic assumptions															
Real GDP growth (in percent)	-1.3	0.6	3.9	2.1	3.1	-3.9	1.3	2.8	2.7	2.5	2.3	1.3	1.7	0.0	1.6
GDP deflator in US dollar terms (change in percent)	1.8	3.3	-4.4	1.3	2.6	2.1	1.5	1.6	1.8	2.0	1.9	1.8	2.0	3.7	2.1
Effective interest rate (percent) 5/	3.5	3.6	3.9	4.5	0.9	3.7	2.9	2.9	2.8	2.7	2.7	2.9	3.5	1.6	2.9
Growth of exports of G&S (US dollar terms, in percent)	1.5	16.7	3.9	5.2	6.1	-2.4	0.5	3.0	3.9	4.1	4.0	2.2	3.7	0.0	3.3
Growth of imports of G&S (US dollar terms, in percent)	-4.1	6.5	3.4	4.2	7.6	-2.0	4.1	4.7	1.7	-0.4	1.8	1.6	3.5	0.2	3.4
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	16.9	38.3	30.7	35.3	31.6	46.4	33.2	37.7	30.6	36.0
Government revenues (excluding grants, in percent of GDP)	36.8	35.7	35.0			40.6	38.3	37.5	37.3	37.4	37.3		36.7	35.2	36.4
Aid flows (in Billions of US dollars) 7/	0.1	0.0	0.0			0.0	0.1	0.1	0.0	0.1	0.1		0.1	0.1	
of which: Grants	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
of which: Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			5.4	8.3	7.8	6.7	6.9	6.3		5.5	4.0	4.8
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			48.5	69.9	62.0	70.7	56.4	69.2		67.3	63.2	65.0
Memorandum items:															
Nominal GDP (Billions of US dollars)	0.5	0.5	0.5			0.5	0.5	0.5	0.6	0.6	0.6		0.7	1.1	
Nominal dollar GDP growth	0.5	3.9	-0.7			-1.9	2.8	4.5	4.5	4.5	4.2	3.1	3.7	3.7	3.7
PV of PPG external debt (in Billions of US dollars)			0.2			0.2	0.2	0.2	0.2	0.2	0.2		0.2	0.3	
(PVt-PVt-1)/GDPt-1 (in percent)						2.6	0.6	1.6	-0.3	2.4	-0.3	1.1	-0.3	1.1	0.2
Gross workers' remittances (Billions of US dollars)	...	...	...			...	...	...	...	...	...		...	...	
PV of PPG external debt (in percent of GDP + remittances)	...	...	38.4			41.8	41.2	41.0	38.9	39.5	37.7		29.8	25.1	
PV of PPG external debt (in percent of exports + remittances)	...	...	92.2			100.9	101.6	102.5	98.0	99.9	95.4		75.5	67.1	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	12.5			15.4	12.5	13.0	12.5	12.5	12.5		11.3	6.7	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table A2. Dominica: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, Active Scenario, 2015-35

	Projections							
	2015	2016	2017	2018	2019	2020	**2025**	2035
PV of debt-to GDP ratio								
Baseline	42	41	41	39	40	38	**30**	25
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	42	43	45	44	44	42	**40**	42
A2. New public sector loans on less favorable terms in 2015-2035 2	42	43	44	44	47	46	**46**	55
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	42	42	43	41	42	40	**32**	27
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	42	42	43	41	41	40	**31**	26
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	42	42	43	41	42	40	**32**	27
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	42	41	41	39	40	38	**30**	25
B5. Combination of B1-B4 using one-half standard deviation shocks	42	40	41	39	40	38	**30**	26
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	42	59	58	55	56	54	**42**	36
PV of debt-to-exports ratio								
Baseline	101	102	102	98	100	95	**75**	67
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	101	107	111	110	110	106	**101**	112
A2. New public sector loans on less favorable terms in 2015-2035 2	101	105	111	110	118	117	**117**	148
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	101	102	102	98	100	95	**75**	67
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	101	104	113	108	110	105	**83**	73
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	101	102	102	98	100	95	**75**	67
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	101	101	103	99	101	96	**76**	68
B5. Combination of B1-B4 using one-half standard deviation shocks	101	96	96	92	94	90	**71**	65
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	101	102	102	98	100	95	**75**	67
PV of debt-to-revenue ratio								
Baseline	103	108	109	104	106	101	**81**	71
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	103	113	119	117	117	112	**109**	119
A2. New public sector loans on less favorable terms in 2015-2035 2	103	112	118	117	124	124	**126**	157
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	103	110	116	111	112	107	**86**	76
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	103	109	114	110	111	106	**85**	73
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	103	111	115	111	112	107	**86**	75
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	103	107	110	105	107	102	**82**	72
B5. Combination of B1-B4 using one-half standard deviation shocks	103	106	110	105	106	101	**81**	73
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	103	153	155	148	150	143	**115**	101

Table A2. Dominica: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, Active Scenario, 2015-35 (concluded)

Debt service-to-exports ratio								
Baseline	15	13	13	13	13	13	**11**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	15	12	13	13	13	13	**13**	10
A2. New public sector loans on less favorable terms in 2015-2035 2	15	13	12	12	11	12	**10**	11
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	15	13	13	13	13	13	**11**	7
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	15	13	14	13	13	13	**12**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	15	13	13	13	13	13	**11**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	15	13	13	13	13	13	**11**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	15	12	13	12	12	12	**11**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	15	13	13	13	13	13	**11**	7
Debt service-to-revenue ratio								
Baseline	16	13	14	13	13	13	**12**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	16	13	14	14	14	14	**14**	11
A2. New public sector loans on less favorable terms in 2015-2035 2	16	13	12	12	12	12	**11**	12
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	16	14	15	14	14	14	**13**	8
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	16	13	14	14	13	13	**13**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	16	14	15	14	14	14	**13**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	16	13	14	13	13	13	**12**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	16	14	15	14	14	14	**13**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	16	19	20	19	19	19	**17**	10
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	26	26	26	26	26	26	**26**	26

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly a
an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table A3. Dominica: Public Sector Debt Sustainability Framework, Active Scenario, 2012-35
(in percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2012	2013	2014	Average 5/	Standard Deviation 5/	2015	2016	2017	2018	2019	2020	2015-20 Average	2025	2035	2021-35 Averag
Public sector debt 1/	72.6	74.7	83.9			85.5	87.7	87.3	85.4	83.4	81.0		72.3	59.1	
of which: foreign-currency denominated	52.6	52.6	53.4			56.6	56.4	55.8	53.2	51.8	50.3		45.8	37.5	
Change in public sector debt	2.8	2.2	9.2			1.6	2.2	-0.4	-1.8	-2.1	-2.4		-1.6	-3.4	
Identified debt-creating flows	5.0	-0.3	4.4			0.1	1.9	-0.8	-2.1	-2.2	-2.5		-2.3	-3.6	
Primary deficit	3.7	0.5	1.6	-2.6	5.7	-4.4	1.9	0.6	-0.6	-0.8	-1.3	-0.8	-2.0	-1.7	-1.1
Revenue and grants	39.5	38.9	37.0			44.8	44.4	43.2	42.5	41.9	41.2		40.6	38.2	
of which: grants	2.7	3.2	2.1			4.2	6.1	5.7	5.2	4.5	3.9		3.9	3.0	
Primary (noninterest) expenditure	43.1	39.4	38.7			40.4	46.3	43.8	41.8	41.1	39.9		38.6	36.5	
Automatic debt dynamics	1.4	-0.8	2.7			4.5	0.0	-1.4	-1.5	-1.5	-1.2		-0.3	-1.9	
Contribution from interest rate/growth differential	1.4	0.1	-0.7			4.4	-0.3	-1.7	-1.5	-1.5	-1.3		-0.3	-1.2	
of which: contribution from average real interest rate	0.5	0.5	2.1			1.1	0.7	0.8	0.7	0.6	0.6		0.9	-0.2	
of which: contribution from real GDP growth	0.9	-0.4	-2.8			3.4	-1.1	-2.4	-2.3	-2.1	-1.8		-1.2	-1.0	
Contribution from real exchange rate depreciation	0.0	-0.9	3.4			0.1	0.3	0.3	0.1	0.0	0.1		0.0	-0.6	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-2.2	2.5	4.8			1.5	0.3	0.3	0.3	0.2	0.1		0.8	0.2	
Other Sustainability Indicators															
PV of public sector debt	...	...	68.9			70.7	72.5	72.5	71.2	71.1	68.4		56.3	46.6	
of which: foreign-currency denominated	...	...	38.4			41.8	41.2	41.0	38.9	39.5	37.7		29.8	25.1	
of which: external	...	...	38.4			41.8	41.2	41.0	38.9	39.5	37.7		29.8	25.1	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	11.0	6.6	9.2			5.4	9.7	10.6	7.7	8.6	8.2		8.0	3.2	
PV of public sector debt-to-revenue and grants ratio (in percent	...	...	186.1			157.8	163.1	167.8	167.7	169.9	166.0		138.7	122.0	
PV of public sector debt-to-revenue ratio (in percent)	...	...	197.2			174.2	189.2	193.1	191.0	190.2	183.1		153.3	132.4	
of which: external 3/	...	...	109.9			103.0	107.6	109.2	104.5	105.7	100.9		81.1	71.2	
Debt service-to-revenue and grants ratio (in percent) 4/	18.6	15.7	20.4			21.8	17.5	23.1	19.6	22.3	23.0		24.8	12.9	
Debt service-to-revenue ratio (in percent) 4/	20.0	17.1	21.6			24.1	20.3	26.6	22.4	24.9	25.4		27.4	13.9	
Primary deficit that stabilizes the debt-to-GDP ratio	0.8	-1.7	-7.5			-6.0	-0.3	1.1	1.2	1.3	1.1		-0.5	1.7	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	-1.3	0.6	3.9	2.1	3.1	-3.9	1.3	2.8	2.7	2.5	2.3	1.3	1.7	1.7	1.7
Average nominal interest rate on forex debt (in percent)	1.6	1.6	2.6	2.5	0.9	2.7	1.5	1.4	1.4	1.2	1.0	1.5	2.1	1.3	1.4
Average real interest rate on domestic debt (in percent)	2.9	2.5	8.6	3.5	2.9	2.5	3.7	3.7	3.4	3.3	3.4	3.3	3.2	0.4	3.0
Real exchange rate depreciation (in percent, + indicates depreci	-0.1	-1.7	6.7	0.7	2.7	0.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	1.8	3.3	-4.4	1.3	2.6	2.1	1.5	1.6	1.8	2.0	1.9	1.8	2.0	3.7	2.1
Growth of real primary spending (deflated by GDP deflator, in p	0.8	-8.2	1.9	-0.5	2.8	0.4	16.2	-2.8	-2.0	0.7	-0.7	2.0	3.2	-3.5	1.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	16.9	38.3	30.7	35.3	31.6	46.4	33.2	37.7	30.6	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

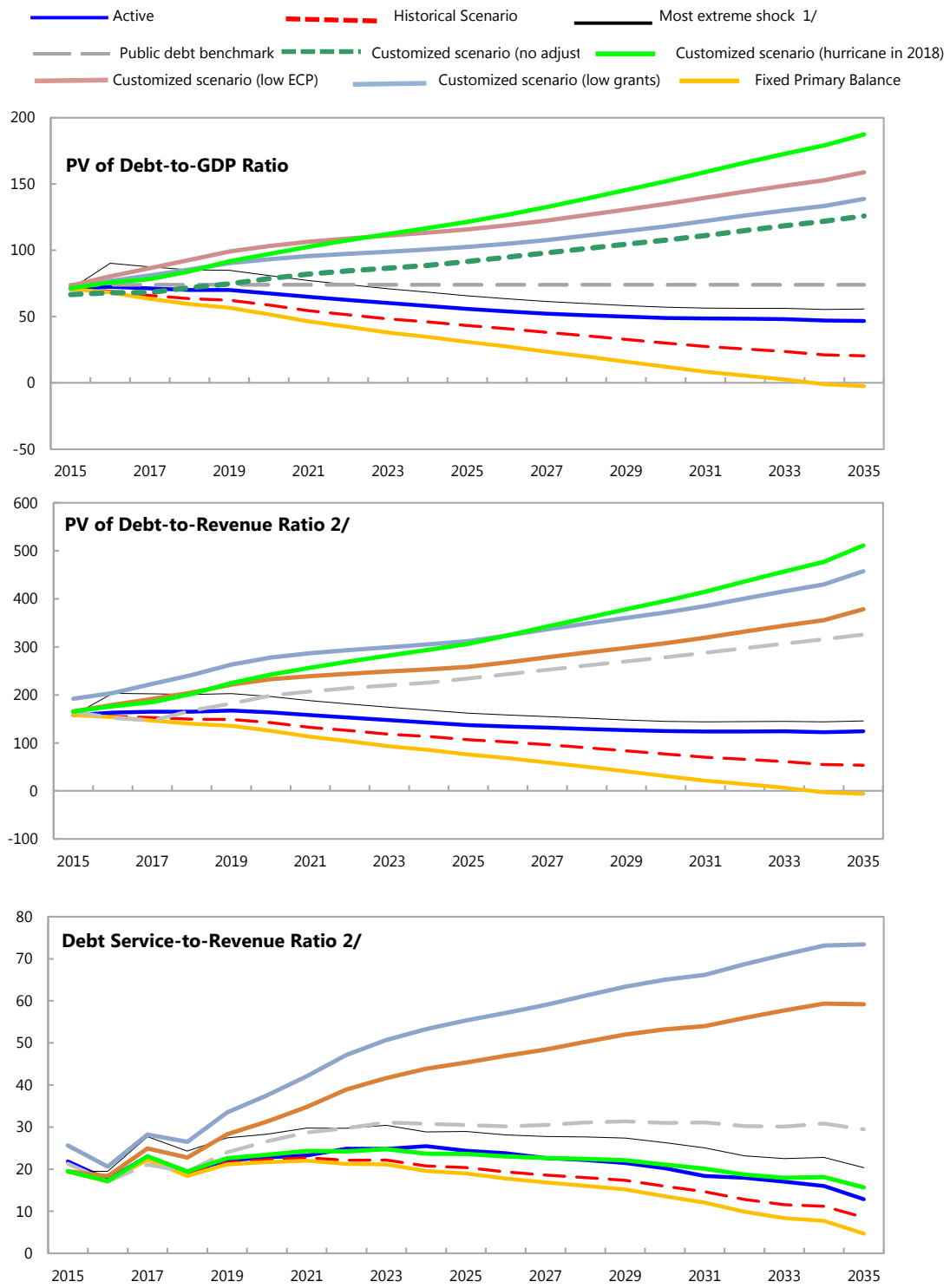
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table A4. Dominica: Sensitivity Analysis for Key Indicators of Public Debt, Active Scenario, 2015-35 (in percent of GDP)

	Projections							
	2015	2016	2017	2018	2019	2020	2025	2035
PV of Debt-to-GDP Ratio								
Baseline	71	72	72	71	71	68	56	47
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	71	69	67	65	64	60	44	21
A2. Primary balance is unchanged from 2015	71	68	65	61	58	53	32	-2
A3. Permanently lower GDP growth 1/	71	73	74	74	75	74	71	91
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	71	75	80	80	81	80	76	79
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	71	74	75	74	74	71	59	48
B3. Combination of B1-B2 using one half standard deviation shocks	71	72	73	73	74	71	63	59
B4. One-time 30 percent real depreciation in 2016	71	90	89	87	86	83	67	56
B5. 10 percent of GDP increase in other debt-creating flows in 2016	71	80	80	79	79	76	64	52
PV of Debt-to-Revenue Ratio 2/								
Baseline	158	163	168	168	170	166	139	124
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	158	155	155	152	152	145	109	55
A2. Primary balance is unchanged from 2015	158	153	149	143	138	128	78	-6
A3. Permanently lower GDP growth 1/	158	165	172	174	179	178	174	237
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	158	168	183	187	193	193	186	205
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	158	166	175	175	177	173	146	127
B3. Combination of B1-B2 using one half standard deviation shocks	158	162	169	171	175	173	156	155
B4. One-time 30 percent real depreciation in 2016	158	203	206	205	207	200	164	146
B5. 10 percent of GDP increase in other debt-creating flows in 2016	158	180	185	185	188	184	157	137
Debt Service-to-Revenue Ratio 2/								
Baseline	22	18	23	20	22	23	25	13
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	19	17	22	19	22	22	21	9
A2. Primary balance is unchanged from 2015	19	17	22	18	21	22	19	5
A3. Permanently lower GDP growth 1/	19	17	23	20	23	24	25	21
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	19	17	24	20	24	25	26	19
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	19	17	23	19	22	23	23	14
B3. Combination of B1-B2 using one half standard deviation shocks	19	17	23	20	23	24	24	16
B4. One-time 30 percent real depreciation in 2016	19	20	28	24	28	28	29	20
B5. 10 percent of GDP increase in other debt-creating flows in 2016	19	17	23	20	23	23	24	15

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Figure A2. Dominica: Indicators of Public Debt Under Alternative Scenarios, 2015-35 (Active Scenario) 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025.
2/ Revenues are defined inclusive of grants.

Annex II. External Sector Assessment

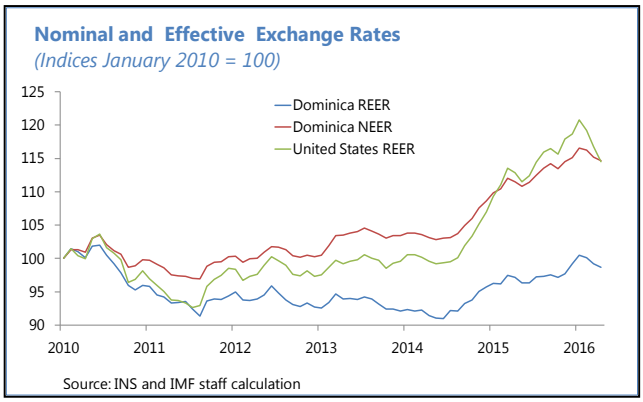
Dominica's external position in 2015 is assessed to be weaker than the level consistent with medium-term fundamentals and desirable policy settings. Achieving a strong and sustainable external sector requires policies and structural reforms over the medium term particularly geared at increasing external competitiveness. Reserves are assessed to be adequate and sufficient to absorb most shocks.

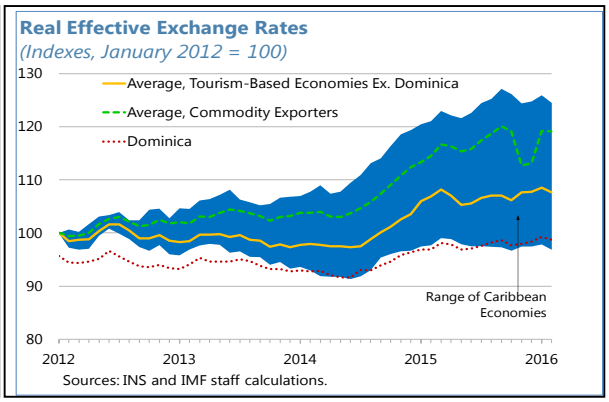
The decline in oil prices, higher than expected tourism receipts and increased inflows of private remittances offset the decline in exports and the increase in imports for reconstruction material following the damages caused by Tropical Storm Erika last August. These developments improved Dominica's current account to a deficit of 9.4 percent of GDP in 2015 from an 11.2 percent of GDP deficit the previous year.[1] Given the forecast for higher imports for reconstruction activity, the REER appreciation and a no-policy change scenario, the current account would worsen in the medium term.

A. Nominal and Real Effective Exchange Rate Assessment

Dominica's nominal effect exchange rate (NEER) appreciated by 7.6 percent on average in 2015 mainly because of the strengthening of the U.S. dollar to which the EC dollar, the regional currency, is pegged. Further appreciation could continue in the near future should relatively stronger growth and higher interest rates in the United States cause the U.S. dollar to strengthen. In comparison the CPI based real effective exchange rate (REER) appreciated by 4.8 percent during the same period and its end-December 2015 index was 99.2 relative to the January 2010 base. Measures of the REER in tourism-competitor Caribbean countries suggest a mild deterioration of about 1 percent in Dominica's competitiveness since June 2015. The recovery in manufacturing and food production following damages sustained from Tropical Storm Erika as well as low energy prices should marginally reduce the headline price level in Dominica in 2016.

[1] The BOP data for 2014 was revised by the Eastern Caribbean Central Bank and Dominica's Statistical Office the National of Trinidad and Tobago (CBTT) to reflect the improved estimation of (i) tourism expenditure, which was previously based on outdated estimates, (ii) transactions by offshore universities and (iii) private remittances. The current account deficit in 2014 has improved from -13.1 percent of GDP to -11.2 percent of GDP. The new data are incomplete. Further improvements in BOP estimation is expected, particularly to FDI and Portfolio flows, as the statistical agencies make progress in BOP compilation using BPM6 methodologies. The revisions could be finalized by the fourth quarter of 2016. CARTAC's technical assistance in 2015 and early 2016 has been instrumental in these developments.





B. External Stability Assessments

Two of the quantitative estimates of Dominica's external position suggest that it is weaker than the level consistent with medium-term fundamentals and desirable policy settings.

- Drawing on the EBA-Lite regression model (after adjusting the results to account for Dominica-specific factors) staff assesses the cyclically-adjusted current account balance and cyclically-adjusted current account norm to be -7.8 percent of GDP and -3.4 percent of GDP, respectively in 2015. The resulting current account gap is -4.4 percent of GDP. Macroeconomic policies contributed -0.2 percent of GDP to the gap. Fiscal policy gaps in other countries contributed -0.9 percent of GDP while reserve adequacy and private credit gaps jointly accounted for 0.8 percent of GDP. The remaining gaps would need to be closed by structural policies to improve competitiveness over the medium term.

- The EBA-Lite methodologies suggest a wide range of Real Effective Exchange Rate (REER) gap estimates. The Current Account regression estimates a REER adjustment of 17.3 percent would be needed to close the current account gap of -4.4 percent of GDP using a standard elasticity. The Index REER regression uses the REER from the INS database and includes policy and fundamental variables that would affect the REER directly or indirectly, through changes to the current account balance indicates. This approach indicates that the REER gap is -1.6 percent (undervalued) and that it is broadly in line with its estimated current account norm.

Current Account and Real Exchange Rate Assessments
(In percent of GDP)

	CA Model	ES Approach	I-REER Model
CA Actual	-9.4	-9.4	
Cyclically Adjusted CA	-7.8	-9.7	
Cyclically Adjusted CA Norm	-3.4	-6.3	
CA Gap	-4.4	-3.4	
Of which: Policy Gap			
Fiscal Policy	*-0.9*		
Private Credit	*0.3*		
Change in Reserves	*0.5*		
Capital Controls	*-0.1*		
Residual	-4.2		
Real Exchange Rate Gap[1]	17.3	13.2	-1.6

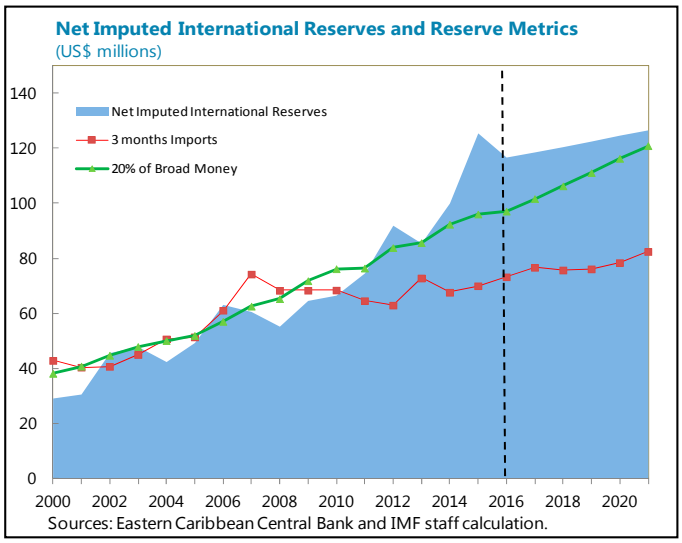
Source: IMF staff estimates.
1/ Data reflect percentage over or under valauation. Positive numbers indicate that REER is overvalued and negative numbers indicate that the REER is undervalued.

- The external sustainability approach sets a medium-term current account norm at the level that would bring net foreign assets-to-GDP to zero in ten years, by end-2025. On this basis, the estimated norm is a deficit of 6.3 percent of GDP, compared to the underlying current account

forecast of a deficit of 8.5 percent of GDP over the 10-year period. A REER adjustment of 13.2 percent would be needed to close the current account gap.

C. Reserve Adequacy

Dominica's reserve position improved in 2015. Dominica is a member of the Eastern Caribbean Currency Union. Under the quasi-currency board arrangement foreign assets and liabilities of the Eastern Caribbean Central Bank (ECCB) cannot be directly assigned to an individual country. The imputed reserves method is used as a proxy for net foreign assets held with the ECCB.[2] At US$125 million, Dominica's imputed reserves at end-2015 covered 5.6 months of imports (forward cover) and 26 percent of broad money exceeding the typical benchmarks of reserve adequacy such as 3 months of imports and 20 percent of broad money, respectively.[3,4] In the active scenario, reserve levels are expected to marginally decline to 4.9 months of imports through 2021 but are adequate for cushioning external shocks and preventing disorderly market conditions.[5]



D. Conclusion

Dominica's external position in 2015 is assessed to be weaker than the level consistent with medium-term fundamentals and desirable policy settings. Notwithstanding the adequate foreign exchange reserves, staff estimates that the external position would weaken over the medium and growth would remain sub-optimal. Fiscal consolidation aiming for a small primary surplus over the medium term while providing space for infrastructure spending to ease supply bottlenecks would improve the current account position and help stabilize reserves. Supportive structural reforms - improving labor productivity, reducing the electricity costs, increasing access to finance as well as reforms to promote the business environment - are also needed to boost productivity, improve competitiveness and prospects for export diversification.

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[2] The computation of the imputed reserves of the individual ECCB member countries are the differences between the assets held by the member countries with the ECCB and the liabilities of the member countries to the ECCB.

[3] IMF 2015, *Assessing Reserve Adequacy –Specific Proposals*. Reserve adequacy assessments for currency unions should consider the reserve needs of the consolidated union level. This should be supplemented by a discussion of factors that have a bearing on the size of reserves, such as the union's financial architecture and supportive institutions, and the correlation of shocks faced by union members.

[4] Information on short term debt and other liabilities is unavailable and consequently assessments against other reserve adequacy metrics such as the IMF's composite Assessment Reserve Adequacy (ARA) metric cannot be computed.

[5] This assessment does not consider the need for additional buffers to deal with significant natural disasters.